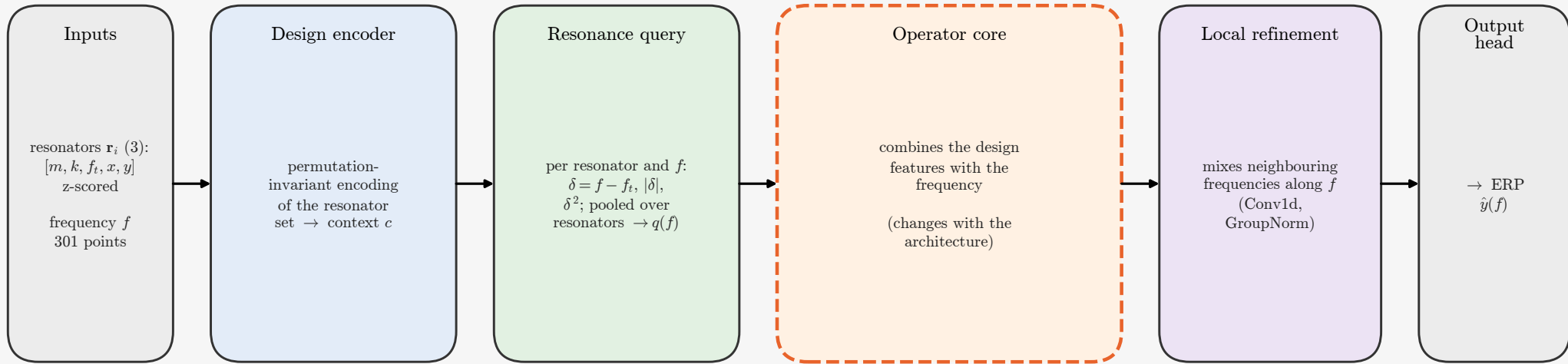


Forward neural operators: common structure

resonator design and frequency $\rightarrow$ ERP spectrum; only the operator core changes between architectures



Which optional blocks (query, refinement) a model uses is listed on its card. The frequency f enters the query and the core.

Operator core of each architecture

NN: plain network encoder: none; flattened $\mathbf{r}_i$ and f enter one MLP core: 6 dense layers (148), ReLU, dropout 0.1 query: no; refinement: no 112,925 parameters	DON: DeepONet encoder: set + sorted (71) core: branch coefficients (4 terms $\times$ 113) $\cdot$ trunk MLP(f); trunk FiLM-modulated by c query: no; refinement: yes 146,294 parameters	DNO: Deep Neural Op. encoder: set + sorted (57) core: 4 FiLM residual blocks (width 54), FiLM from the context and query query: yes; refinement: yes 149,994 parameters	FNO: Fourier NO encoder: set + sorted (23) core: 4 Fourier blocks, 35 modes (spectral + local conv), frequency axis padded by 8 query: yes; refinement: in blocks 153,564 parameters	DCO: Deep Cat Operator encoder: set + sorted (56+56) core: concatenate branch, trunk MLP(f) and query (196) $\rightarrow$ lift $\rightarrow$ 4 residual blocks (67) query: yes; refinement: yes 155,240 parameters
GNO: Graph NO encoder: resonators as graph nodes, 3 message-passing layers core: kernel MLP on node, f , δ ; attention over resonators refinement: yes 112,230 parameters	STO: Set Transformer Op. encoder: resonator tokens, 2 Transformer layers (4 heads) core: cross-attention from frequencies to tokens (detuning bias); own frequency mixer 110,193 parameters	SIREN: SIREN NO encoder: set + sorted (57) core: 4 sine layers ($\omega_0 = 20$) on $[f, q(f)]$, FiLM-modulated by the context c query: yes; refinement: yes 152,746 parameters	WNO: Wavelet NO encoder: set + sorted (57) core: 4 wavelet blocks (3-level Haar transform, learned mixing per level) query: yes; refinement: in blocks 145,946 parameters	LNO: Laplace NO encoder: set + sorted (90) core: 13 poles and residues predicted from c ; features $r/(s-p) + \bar{r}/(s-\bar{p})$, $s = jf$ query: yes; refinement: yes 162,704 parameters

Set encoder: shared MLP, mean + max pooling; sorted encoder: resonators ordered by f_t , concatenated, MLP; each resonator carries 120 plate-mode features. Numbers in brackets are widths.

GNO and STO have no set / sorted encoder: their resonators are graph nodes or attention tokens. ERP values are normalised before the loss.

Loss functions

